

Shiying Xiao

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Education

- **M.Sc. in Statistics** August 2023 – May 2026
University of Connecticut
Advisor: [Prof. Jun Yan](#) Storrs, CT
- **M.Econ. in Statistics** September 2019 – July 2022
Shanxi University of Finance and Economics
Advisor: [Prof. Tao Wang](#) Taiyuan, China
- **B.Econ. in Insurance** September 2015 – July 2019
Shanxi University of Finance and Economics
Taiyuan, China

Research Interests

Applied Econometrics, Biostatistics, High-Dimensional Data, Input-Output Analysis, National Economic Accounting, Network Analytics, Public Health Applications, Statistical Computing

Publications (* refers to the corresponding author)

1. Thornton, S., Szymanski, M. R., Brewer, G. J., Casa, D. J., Moore, T. E., **Xiao, S.**, Green, A. A., Pardee, K. and Lee, E. C.* (2026). Sample processing methods affect salivary metabolomics in human exercise-stress studies. *Metabolomics*, 22(4): 123. [DOI](#)
2. Seyednejad, A., Sacko, T. J., Babigian, C. J., Moore, T. E., **Xiao, S.**, Liddle, J. C. and Sartor, G. C.* (2026). Cocaine-induced changes in the BRD4 interactome identifies casein kinase 1 epsilon as a therapeutic target. *Journal of Neurochemistry*, 170(1): e70336. [DOI](#)
3. Wang, T.* and **Xiao, S.** (2022). A comparative study of domestic industrial circulation between China and the United States from the perspective of input-output network. *Statistical Research*, 39(11): 32-43. [URL](#) (in Chinese with English abstract)
4. **Xiao, S.***, Yan, J. and Zhang, P. (2022). Incorporating auxiliary information in betweenness measure for input-output networks. *Physica A: Statistical Mechanics and its Applications*, 607: 128200. [DOI](#)
5. Wang, T., **Xiao, S.**, Yan, J. and Zhang, P.* (2021). Regional and sectoral structures of the Chinese economy: A network perspective from multi-regional input-output tables. *Physica A: Statistical Mechanics and its Applications*, 581: 126196. [DOI](#)

Preprints and Manuscripts

1. Aladelokun, O., Benitez, K., Shen, X., Jain, A., Berardi, D., Lu, L., Gardner, E., Yang, J., Almazwini, M., Rattray, N. J. W., Butensky, S., **Xiao, S.**, Siddique, S., Berner, A. M., Aalizadeh, R., Vasiliou, V., Paty, P. B., Garcia-Milian, R., Khan, S. A. and Johnson, C. H.* (2026+). Sex-specific metabolic phenotypes in early onset colorectal cancer. Under review.
2. Chen, C.*, **Xiao, S.** and Salike, N. (2026+). China's export supply chain resilience measures and improvement strategy: The application of product space theory. (in Chinese with English abstract)
3. Wang, T., **Xiao, S.*** and Yan, J. (2026+). Comparison of sectoral structures between China and Japan: A network perspective. [arXiv](#). Under review.
4. **Xiao, S.***, Yan, J. and Zhang, P. (2026+). Groupwise regularized adaptive sparse precision solution.
5. **Xiao, S.**, Liddle, J., Balsbaugh, J., Watt, C. and Moore, T.* (2026+). msDiaLogue: Analyzing and visualizing DIA proteomics data in R.

6. Zhang, P.*, **Xiao, S.**, Robb, W. H., Liu, D., Jefferson, A. L. and Yan, J. (2026+). Gaussian graphical models for functional connectivity analysis: A statistical review with applications to Alzheimer’s disease. [arXiv](#). Under review.

Presentations and Posters

1. Functional Connectivity Estimation Based on Gaussian Graphical Models: A Statistical Review and Applications to Alzheimer’s Disease Data. Datablitz speaker. The 27th Annual Neuroscience at Storrs Symposium. University of Connecticut, Storrs, CT, 10/22/2024.
2. Gaussian Graphical Models for Functional Connectivity Analysis: A Statistical Review and Applications to Alzheimer’s Disease Data. Poster presenter. The 33rd Annual Applied Statistics Symposium for the International Chinese Statistical Association (ICSA 2024). Nashville, TN, 06/17/2024.
3. Functional Connectivity Analysis in Brain Networks: A Statistical Review and Application to Alzheimer’s Disease Data. Invited speaker. [The 37th New England Statistics Symposium \(NESS 2024\)](#). University of Connecticut, Storrs, CT, 05/23/2024.
4. Introduction to Data Analysis in R for DIA-based Proteomics Using msDiaLogue. Assistant instructor. [The 4th Annual Introduction to Mass Spectrometry-Based Proteomics Workshop](#). University of Connecticut, Storrs, CT, 03/15/2024.

Software

1. **Xiao, S.**, Yan, J. and Zhang, P. (2026). **grasps**: Groupwise Regularized Adaptive Sparse Precision Solution. R package version 0.1.1, <https://CRAN.R-project.org/package=grasps>.
2. **Xiao, S.**, Yan, J. and Zhang, P. (2026). **spice**: Sparse Precision (Inverse Covariance) Estimation. R package version 0.1.0, <https://shiyang-xiao.com/spice>.
3. **Xiao, S.**, Moore, T. and Watt, C. (2025). **msDiaLogue**: Analysis and Visuals for Data-Independent Acquisition Mass Spectrometry Data. R package version 0.0.7, <https://uconn-scs.github.io/msDiaLogue>.
4. **Xiao, S.** (2025). **pecan**: Portfolio for Economic Complexity Analysis and Navigation. R package version 0.1.0, <https://CRAN.R-project.org/package=pecan>.
5. **Xiao, S.**, Yan, J. and Zhang, P. (2024). **ionet**: Network Analysis for Input-Output Tables. R package version 0.2.2, <https://CRAN.R-project.org/package=ionet>.

Experience

Statistical Consultant

August 2023 – Present

UConn Statistical Consulting Services

Storrs, CT

- Developed **msDiaLogue**, an R package providing reproducible statistical workflows, publication-quality visualizations, and analytical tools for untargeted mass spectrometry-based proteomics data analysis.
- Provided statistical consulting, data analysis, and reproducible programming support to researchers across diverse academic disciplines.
- Contributed statistical analyses to multiple interdisciplinary collaborative research projects, resulting in co-authorship on three publications.

Research Assistant

August 2023 – May 2026

University of Connecticut

Storrs, CT

- Conducted research on precision matrix estimation for brain functional connectivity analysis using resting-state fMRI data, with applications to Alzheimer’s disease research.
- Conducted a comprehensive benchmarking study of existing methods for precision matrix estimation and developed the accompanying R package **spice** to provide a unified and reproducible implementation of existing precision matrix estimation methods.
- Developed a novel sparse-group regularized method for precision matrix estimation with predefined group structures and implemented the method in the accompanying R package **grasps**.
- Evaluated the proposed method through large-scale simulation studies on high-performance computing (HPC) systems and applications to resting-state fMRI data.

Assistant Professor of Teaching

Fuzhou University of International Studies and Trade

August 2022 – July 2023

Fuzhou, China

- Selected Teaching Activities

1. Data Visualization Design (C010405620): 2 credit undergraduate level; Spring 2023.
2. Business Data Mining & Analytics (C010402930): 2 credit undergraduate level; Spring 2023.

- Research Program

The National Social Science Fund of China, 23CGJ018, Research on the Reconstruction of China-ASEAN Foreign Trade Industry Value Chain and Resilience Enhancement Strategies. PI: Chen Chen.

Research Assistant

Shanxi University of Finance and Economics

June 2019 – July 2022

Taiyuan, China

- Projects in the area of network analysis and complex systems: development of methodology, management activities to maintain research data, implementation of the computer code, and data presentation.

1. The National Social Science Fund of China, 21BTJ013, Research on the Compilation and Application of the International Flow of Funds Table. PI: Tao Wang.
2. National Bureau of Statistics of China, 2018LZ33, Research on the Compilation and Model Application of Inter-Provincial Input-Output Tables in China. PI: Tao Wang.

- Projects in the area of evaluation index system: ideas, data collection and cleaning, application of statistical to analyze data, and visualization.

1. Shanxi Provincial Federation of Social Sciences, SSKLZDKT20211075, Study on the Index System of High-Quality Tourism Development in Shanxi Province.
2. Shanxi Academy of Social Sciences, 2020YY119, Study on the Index System of High-Quality Development in Shanxi.
3. Shanxi Provincial Bureau of Statistics, KY[2020]046, Study on the Scale Benefit of High-Tech Industry in Shanxi.
4. Shanxi Provincial Bureau of Statistics, KY[2019]151, Study on the Evaluation Index System of Industrial Competitiveness in Shanxi Province.
5. Shanxi Academy of Social Sciences, 2019B100, Study on the Monitoring and Evaluation Index System of Rural Social Undertakings in Shanxi Province.

Reviewer Service

- Journal of Open Source Software
- Physica A: Statistical Mechanics and its Applications

Technical Skills

- **Programming:** R, Python, C++
- **Computing:** Linux, Bash, High-Performance Computing (HPC)
- **Tools:** Git, L^AT_EX, Markdown, Quarto